Dipartimento di Ingegneria Informatica, Automatica
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Parameter-Efficient Few-Step Distillation of Diffusion Models via LoRA

Computer Vision — Project 5

Lightweight Diffusion Models for Resource-Constrained Environments

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1 Introduction

Denoising Diffusion Probabilistic Models (DDPMs) [1] produce high-quality images, but their sampling procedure is an iterative Markov chain of hundreds to thousands of network evaluations, which makes inference slow and expensive. This is a critical limitation for deployment on edge devices or for training in resource-constrained environments.

This project, following the Project 5 specification, implements a standard DDPM baseline on CIFAR-10 and integrates a methodological novelty aimed at accelerating sampling. We then rigorously evaluate the trade-off between computational efficiency (sampling time) and generation quality (FID, Inception Score) as well as sample diversity (Precision/Recall).

Deterministic samplers such as DDIM [2] reduce the number of steps, but quality collapses once the step count becomes very small. The root cause is the *accumulated discretization error* along the reverse trajectory, not a failure of noise prediction at individual timesteps. Our contribution directly targets this error while keeping the adaptation cheap:

> **Parameter-efficient few-step distillation via LoRA.** We freeze the trained DDPM (the *teacher*) and train only low-rank adapters [4] injected into the attention layers (the *student*) with a progressive-distillation-style trajectory-matching objective [3]. The student learns to reproduce, in very few steps, what the teacher produces with many DDIM sub-steps.

Because only the adapters are trained (a small fraction of the weights) and they can be merged into the base weights at inference, the method adds essentially no inference overhead while providing large speed-ups over the full-step baseline.

2 Related Work and Identified Limitations

- **DDPM** [1] — our baseline generative model. *Limitation:* very slow sampling ($T = 1000$ sequential evaluations).
- **DDIM** [2] — deterministic non-Markovian sampler enabling fewer steps. *Limitation:* quality collapses under aggressive step reduction (in our measurements, FID rises from $\sim$29 at 100 steps to $>$100 at 4 steps).
- **Progressive Distillation** [3] — distils a teacher into a student that halves the number of steps. *Limitation:* requires full-model fine-tuning at each stage, costly on constrained hardware.
- **LoRA** [4] — low-rank adapters for parameter-efficient adaptation.

Our novelty sits at the intersection: we bring progressive-distillation-style trajectory matching to diffusion *but perform it with LoRA only*, addressing the cost limitation of distillation while attacking the discretization-error limitation of few-step DDIM.

3 Method

3.1 Baseline DDPM

We use an attention U-Net (encoder–decoder with skip connections, GroupNorm, self-attention at low resolution, sinusoidal timestep embeddings). The forward diffusion process noises a clean image x_0 as

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{1}$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s)$ follows a linear β-schedule with $T = 1000$. The network ε_θ is trained with the standard noise-prediction objective

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{x_0,\,t,\,\varepsilon}\big\|\varepsilon - \varepsilon_\theta(x_t, t)\big\|_2^2, \tag{2}$$

using an exponential moving average (EMA) of the weights and mixed precision (AMP).

3.2 Fast Sampling (DDIM)

Deterministic DDIM ($\eta = 0$) advances from timestep t to t' via

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\,\varepsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}, \qquad x_{t'} = \sqrt{\bar{\alpha}_{t'}}\,\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{t'}}\,\varepsilon_\theta(x_t, t). \tag{3}$$

Reducing the number of visited timesteps accelerates sampling but increases the discretization error between consecutive $x_{t'}$ estimates.

3.3 Novelty: LoRA Few-Step Distillation

The trained baseline (EMA weights) is the frozen *teacher* ε_ϕ. The *student* ε_θ is the same network with LoRA adapters injected into the attention projections; only the adapters are trainable.

For a target step budget K we build a short schedule and, for a randomly sampled interval $(t \to t')$, we forward-diffuse a real image to x_t via Eq. (1). The teacher refines x_t from t to t' using m small DDIM sub-steps (Eq. (3)), producing the target $x_{t'}^{\text{teacher}}$; the student must reach the same point in a *single* DDIM step, producing $x_{t'}^{\text{student}}$. The objective is a trajectory-matching loss with an auxiliary noise-consistency term:

$$\mathcal{L}_{\text{distill}} = \big\|x_{t'}^{\text{student}} - x_{t'}^{\text{teacher}}\big\|_2^2 \;+\; \lambda\,\big\|\varepsilon_\theta(x_t, t) - \varepsilon_\phi(x_t, t)\big\|_2^2. \tag{4}$$

With $m = 2$ this reduces to canonical progressive distillation. We train one student per target $K \in \{8, 4, 2, 1\}$. At inference the adapters can be merged into the base weights, so the distilled model costs the same as the base model at the same K.

4 Experimental Setup

- **Data:** CIFAR-10 (32×32 RGB); metrics computed against the CIFAR-10 test split via `torch-fidelity`.
- **Model:** U-Net with `base_channels`$= 64$, `time_dim`$= 256$ (single-GPU scale).
- **Training:** 80 epochs, batch size 128, AdamW, EMA, AMP.
- **Distillation:** 10 epochs per student, LoRA on the attention projections (rank 4 by default; ranks 2 and 8 for the ablation).
- **Metrics:** FID, KID, Inception Score (fidelity); Precision/Recall (diversity) [5]; seconds per image (speed). All computed on 10 000 generated images from a shared, seeded noise bank.
- **Reproducibility:** fixed seed (42), deterministic algorithms, seeded `torch.Generator`, pinned package versions.

5 Results

5.1 Baseline: quality and speed vs. number of steps

Table 1 reports the baseline across step budgets. Fewer steps means faster but lower-quality sampling; crucially, **recall (diversity) collapses** long before precision does.

Table 1: Baseline: DDIM at different step counts (FID on 10 000 images). The DDPM row uses 1000 images and is reported only for speed-up factors.

Sampler	Steps	FID ↓	IS ↑	Precision	Recall	s/img	Speed-up†
DDPM (ref)	1000	67.3*	6.26	0.847	0.381	0.678	1×
DDIM	100	29.33	7.23	0.751	0.326	0.053	12.8×
DDIM	50	29.79	7.32	0.777	0.306	0.027	25.2×
DDIM	20	32.43	7.32	0.834	0.243	0.011	60.5×
DDIM	10	40.89	7.02	0.893	0.141	0.006	112.9×
DDIM	8	47.12	6.83	0.903	0.102	0.005	136.5×
DDIM	4	103.28	5.29	0.853	0.015	0.003	242.4×
DDIM	2	188.84	3.73	0.401	0.001	0.002	∼360×
DDIM	1	325.11	1.56	0.005	0.000	0.001	∼510×

† vs. DDPM-1000. * On 1000 images (noisier).

Fidelity vs. Diversity. As steps drop from 100 to 8, precision actually *rises* ($0.75 \to 0.90$) while recall *collapses* ($0.33 \to 0.10$, and to ~ 0 at 4 steps and below). Few-step sampling does not primarily degrade individual samples — it shrinks the portion of the data distribution the model covers. FID alone would hide this; Precision/Recall makes it explicit.

5.2 Novelty: matched-step comparison

Table 2 compares plain DDIM-K against the distilled LoRA-K at the same step budget, same initial noise, and same base weights — so the only difference is the LoRA adapter.

Table 2: Matched step budget K: plain DDIM vs. distilled LoRA.

K	DDIM FID	Distilled FID	ΔFID	DDIM Recall	Distilled Recall	Time ratio
8	47.12	**38.24**	**+8.9**	0.102	0.159	1.01×
4	103.28	**68.34**	**+34.9**	0.015	0.051	1.00×
2	188.84	**159.12**	**+29.7**	0.001	0.002	1.01×
1	325.11	326.57	−1.5	0.000	0.000	1.00×

The distilled student improves FID at $K = 8, 4, 2$, with the largest gain at $K = 4$ (+34.9), and partially **restores diversity** (recall +0.057 at $K = 8$). At $K = 1$ both models fail. Inference cost is unchanged (time ratio ≈ 1): the distilled $K = 8$ model is $\sim$10.6× faster than DDIM-100 and the $K = 4$ model $\sim$18.9× faster, at a fraction of the quality loss of plain DDIM at the same step count.

5.3 Ablation 1 (causal): distillation objective vs. naive eps-loss

To prove the gain stems from the distillation *objective* and not from the added LoRA capacity, we train an identical LoRA (same rank, same trainable parameters) with the standard noise-prediction loss (Eq. (2)) at low steps.

Table 3: Causal control: naive eps-loss LoRA vs. distillation, matched K and rank.

K	Naive-eps LoRA ΔFID	Distillation ΔFID
8	+0.96	**+8.9**
4	−3.51 (worse)	**+34.9**

The naive control yields $\approx$0 improvement at $K = 8$ and is *worse* than plain DDIM at $K = 4$, while distillation yields large gains. This isolates the trajectory-matching objective as the direct cause of the improvement.

5.4 Ablation 2: LoRA rank ($K = 8$)

FID: r=2: 38.32, r=4: 38.48, r=8: 37.67. Rank barely matters — even rank 2 captures almost all of the benefit, reinforcing the parameter-efficient claim.

5.5 Ablation 3: distillation state source ($K = 8$)

Forward-diffused states only: 38.48; mixing teacher-trajectory states ($p = 0.5$): 38.24 — essentially tied. Conditioning on generated trajectory states does *not* help; forward-diffusion states are sufficient. We report this as a negative result and do not rely on trajectory-state mixing in the final method.

6 Limitations

- **Weak absolute baseline.** Under the single-GPU budget the baseline reaches FID $\approx$ 29; samples show global structure but limited detail. Because all comparisons are relative to the same teacher, the conclusions about the novelty hold regardless of absolute quality.
- **Breakdown at $K = 1$.** The method does not recover single-step sampling on this baseline.
- **FID sample-size sensitivity.** We evaluate all headline comparisons at a fixed 10 000 images. The $K = 4$ result (+34.9) is well beyond this noise; the $K = 8$ result (+8.9) is smaller and would benefit from multiple seeds.
- **Linear β-schedule.** A cosine schedule could improve baseline quality and is left as future work.

7 Conclusion

We presented a parameter-efficient few-step distillation of a DDPM using LoRA adapters and a trajectory-matching objective. On CIFAR-10 the method improves few-step generation quality at a matched step budget (up to +34.9 FID at 4 steps), partially restores the diversity lost by aggressive step reduction, and adds no inference overhead, yielding $\sim$10–19$\times$ speed-ups over the DDIM-100 baseline (over 100$\times$ vs. full DDPM). A naive-eps control confirms that the distillation objective — not the extra parameters — is responsible for the gain. Future work includes a cosine schedule, min-SNR-weighted distillation, and rank scaling for the extreme $K = 2/1$ regime.

References

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